

High-resolution 3D Reconstruction with Neural Mesh Shading

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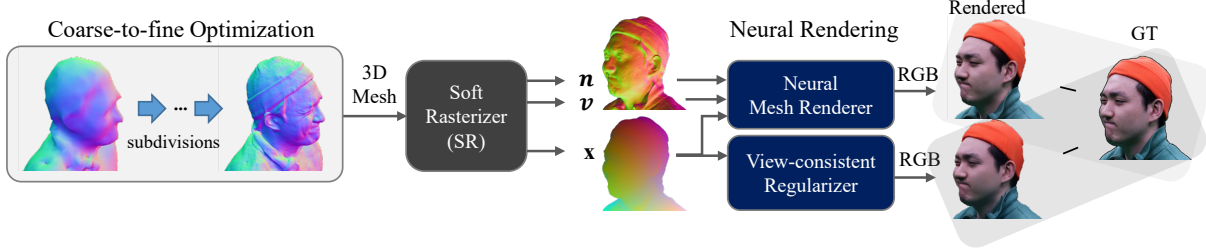


Figure 1: The overview of neural mesh shading framework. From an initial mesh, the 3D mesh vertices and a neural mesh renderer are jointly optimized to represent the target images through the neural mesh renderer. A differentiable soft rasterizer is used to render 3D meshes.

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1 INTRODUCTION

Accurate reconstruction of 3D objects using multiple images has been an essential problem in computer vision and computer graphics. Traditional methods used a consecutive pipeline that finds matched pixels between paired images to estimate disparity, converts 3D depths or points from the disparity, and fuses the accumulated 3D points obtained from multiple depths to generate mesh surfaces. Although the traditional pipeline has shown impressive reconstruction results, the final reconstructed geometry is prone to errors that accumulate at each pipeline step. To address accumulated errors and achieve high-quality reconstructions, many recent studies [Wang et al. 2021; Yariv et al. 2020] directly reconstructed 3D surfaces from multiview images by learning implicit neural representation without the supervision of 3D geometry. Specifically, the surface representation of signed distance functions (SDFs), modeled by a multilayer perceptron (MLP), is used to implicitly represent object surfaces, as described in Fig. 2. In addition, another MLP is used to represent the target's appearance with 3D geometry and viewing information, such as 3D position x , surface normal n , and viewing direction v . However, training two separate networks

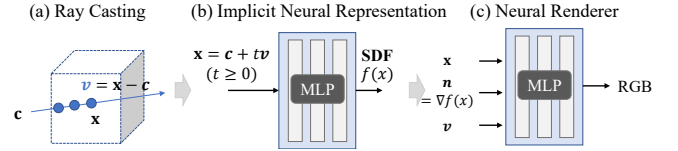


Figure 2: Brief architecture view of the existing reconstruction techniques using implicit surface representation.

with respect to geometry and appearance requires sufficient computational power and space, making it inefficient for high-resolution reconstruction in terms of time and memory.

To address the problem, we propose a high-resolution 3D reconstruction method with neural mesh shading (NMS), an accurate and fast mesh reconstruction method. For given multiview images, an initial 3D mesh is constructed using object masks with a shape from the silhouette technique and optimized to accurately represent the target with a neural mesh renderer. Meanwhile, the neural mesh renderer is optimized to accurately render target images using x , n , and v , as described in Fig. 1. We use a soft rasterizer (SR) [Liu et al. 2020] to obtain geometry information from 3D meshes in the image coordinates. To ensure that rendered colors are consistent across multiple viewpoints, we also introduce a view-consistent regularizer modeled with a shallow MLP. The 3D mesh representation enables much faster optimization and little memory consumption compared to implicit representation, while the neural mesh renderer learns the target appearance from stable surface information from SR. Through comparisons, the experimental results demonstrate that our method outperforms the previous methods qualitatively and quantitatively and provides a strong benefit over other methods in generating high-resolution 3D geometry in less time.

2 OUR APPROACH

Architecture. Similarly to the work of NeRF [Mildenhall et al. 2021], positional encoded 3D location $\gamma(x)$ is passed through 10 fully connected layers for the neural mesh renderer. A skip connection is concatenated into the activation of the fifth layer, where $\gamma(\cdot)$

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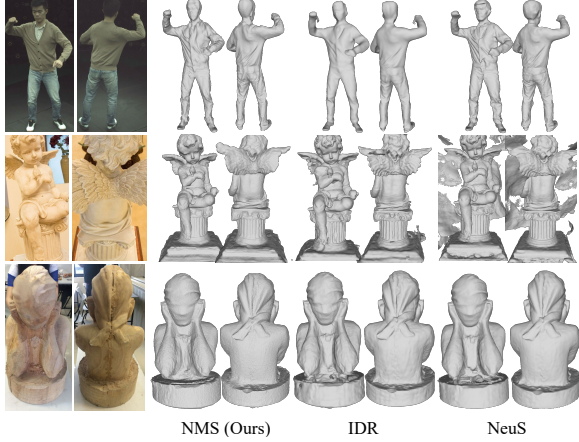


Figure 3: Qualitative comparisons of multiview 3D surface reconstructions on BlendedMVS (©Yao Yao) and ZJU-MoCap (©Sida Peng) datasets. ©Joe Smith.

Table 1: Quantitative comparisons on BlendedMVS dataset.

Method	IDR	NeuS	NMS (Ours)
Chamfer l_1	0.61	0.48	0.38
PSNR (%)	23.55	28.24	30.04
LPIPS	0.122	0.101	0.051
Time (min)	522.12	362.45	4.24

denotes the positional encoding function. The surface normal n and the viewing direction v are concatenated to the activation of the ninth layer. A final layer outputs the rendered RGB value at position x in direction v . The view-consistent regularizer network also produces the RGB value from the input 3D location. We used a similar but shallow architecture, *i.e.*, four fully connected layers, for the regularizer. Furthermore, neither n nor v is used as input to the view-consistent regularizer so that it penalizes the inconsistency of the view between 3D locations x and the rendered values.

Loss. Our objective function is divided into shading and geometry terms as $\mathcal{L} = \mathcal{L}_{\text{shading}} + \alpha \mathcal{L}_{\text{geometry}}$ where α is a balancing parameter. The shading loss is defined by comparing the estimates from the neural mesh renderer with the image values as $\mathcal{L}_{\text{shading}} = \mathcal{L}_{\text{color}} + \beta_1 \mathcal{L}_{\text{silhouette}} + \beta_2 \mathcal{L}_{\text{view}}$. We use the definition in [Liu et al. 2020] for silhouette loss $\mathcal{L}_{\text{silhouette}}$. Like other methods [Wang et al. 2021; Yariv et al. 2020], the color term $\mathcal{L}_{\text{color}}$ is defined with l_1 distance between the color rendered by the neural mesh renderer and the value of the image. Similarly, the view-consistency term $\mathcal{L}_{\text{view}}$ is defined with l_1 distance between the color rendered by the regularizer network and the image value. Similarly to total variation and eikonal losses in SDF-based methods [Wang et al. 2021; Yariv et al. 2020] to regularize grid values, we used general geometry regularization terms to obtain smooth mesh surfaces. The geometry regularization loss is defined with Laplacian and normal consistency terms as $\mathcal{L}_{\text{geometry}} = \mathcal{L}_{\text{laplacian}} + \gamma \mathcal{L}_{\text{normal}}$.

3 EXPERIMENTS

Implementation Details. We trained our network using Adam with initial learning rates of $1e-4$ for the mesh vertices and the neural mesh renderer. Each time the mesh is upsampled by the subdivision method, the learning rate of the mesh vertices is halved.

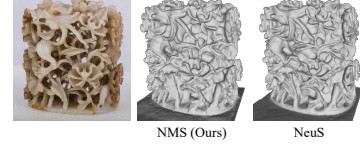


Figure 4: Reconstruction on BlendedMVS (©Yao Yao) dataset.

Experiments were carried out with a single *Nvidia 3090* GPU, using the *PyTorch* framework. Furthermore, we use the *PyTorch3D* framework and the official implementation of SR for our framework.

Evaluation. To evaluate our method, we performed both quantitative evaluations on high-resolution images (2048×1536) of the BlendedMVS dataset [Yao et al. 2020], a large collection of high-resolution multiview images of 113 scenes. For practical reasons, we used the first 15 image sets for evaluation. Furthermore, we used the ZJU-MoCap dataset [Peng et al. 2021] to evaluate the multiview reconstruction of the clothed human body. We compared our method with recent neural reconstruction techniques: IDR [Yariv et al. 2020] and NeuS [Wang et al. 2021]. For fair comparisons, we made the reconstructed meshes from the methods have a similar number of vertices and faces. A quantitative comparison is summarized in Table 1 in terms of Chamfer distance l_1 , peak signal-to-noise ratio (PSNR), and the learned perceptual image patch similarity (LPIPS). In addition, qualitative comparisons are shown in Fig. 3.

Discussion. We proposed a high-resolution 3D reconstruction method with neural mesh shading. Compared to the previous approach, our method significantly benefits in reconstructing high-quality 3D geometry in less time. As our method directly optimizes the mesh vertices, insufficient depths are obtained when reconstructing largely concave objects, as described in Fig. 4. We hope to extend our approach to moving objects with non-rigid registration.

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